TELCO Churn-DATA ANALYTICS PROJECT

TELCO CHURN DATA QUESTIONS

- Which customers are most likely to churn?
- How does tenure affect churn?
- Which contract types and payment methods drive churn.
- How do monthly charges influence churn?
- How much revenue is made monthly and the total revenue?

DATA SOURCE

- **Is the data valid, credible, accurate and is it from a trusted data source?**
  The data is from Kaggle and it is not biased in any way and it has been cleaned and verified to avoid/prevent any errors during the analysis process and also so that the insights are accurate and correct.

DATA CLEANING

- Data was Cleaned in both Microsoft excel and Python by checking if there are duplicates in the dataset, removing null records, fixing typos and also checking the validity of data in the dataset and removing data that is not valid.

LIMITATIONS AND PROBLEMS

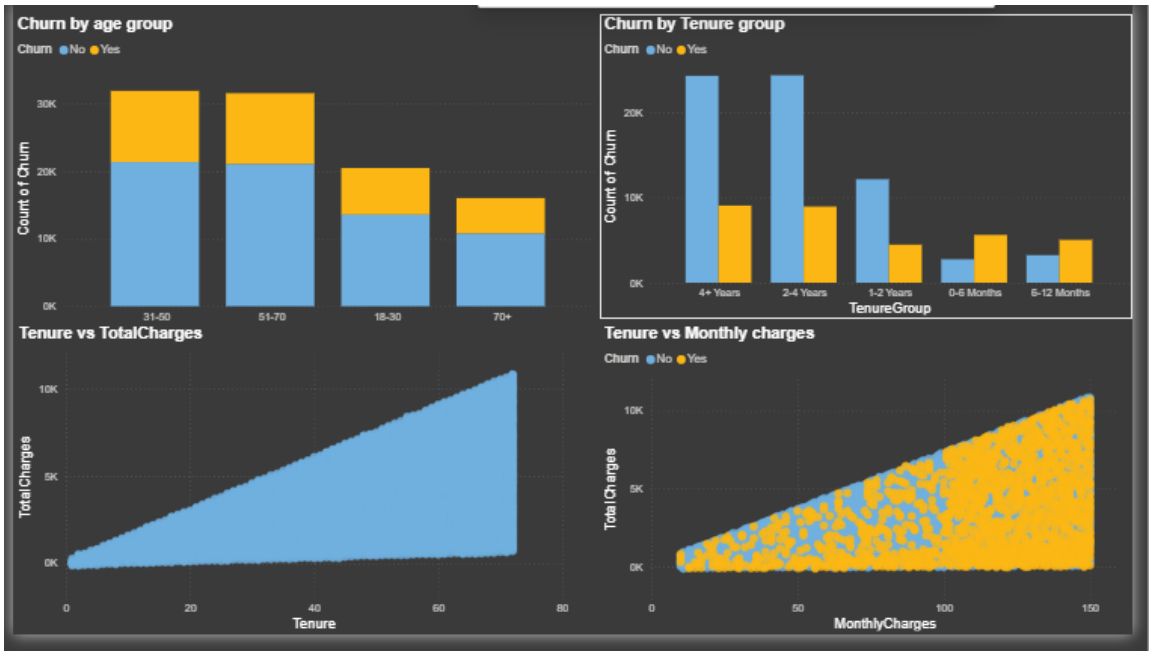
- Due to my process of learning data scientist machine learning and predictions, I ran into a problem whereby the predictions for “will churn” and “will not churn” gave me 0% both which requires me to go back to the drawing board and check where I did the mistakes.

FINDINGS AND INSIGHTS

Upon processing and analysing the data about TELCO churn, I discovered the following:

- $8M is made in monthly charges
- $293M is made in monthly charges
- The average tenure is 36,53 months.
- 33144 people stop (churn) their subscriptions at TELCO while 66856 people don't (not churn) and this indicates that 1/3 people unsubscribe or cancel their subscriptions
- Electronic check is the most used payment method for customers at TELCO and most customers who churn use this type of payment method and the rate for churn on this payment method is slightly 1/3 as well as for other payment methods.
- Month to month contract type is the most used contract and 29348 people do not cancel their subscription while 25 567 people cancel or leave their subscriptions.
    - One-year contract type is second most used contract type while 21 030 on this contract type do not leave or cancel their subscriptions and 4231 people cancel their subscriptions
    - Two-year contract type is second most used contract type while 16 478 on this contract type do not leave or cancel their subscriptions and 3346 people cancel their subscriptions
    - Insight (My assumption) – It seems that month to month subscribers which may be new subscribers are the most users/people who churn/ their churn rate is higher than other contracts type users. One- and two-years contract

types users have a low churn rate which may suggest that these are loyal customers who don't or have least complaints about Telco services.



- **Churn by age group:** Telco have more customers between the age of 30-70 and while that being the case, users from the age 30-70 have the highest number of people who cancel their contracts. This being the case, I also noticed that the churn rate for most age groups/categories is around 50%.
- **Churn by tenure group:** Users with tenure between 0-12 months have a high rate of churn while users with 1 to +4 years of tenure have a low churn rate. This supports my insight that new customers/subscribers have the high churn rate than loyal users/subscribers who have a tenure of 1 to 4 years plus.
- **Tenure vs Monthly charges:** On the scatter plot chart, the yellow dots (churn = yes) dominate the higher monthly-charge range which indicates that customers with higher monthly fees are more likely to churn while blue dots (non-churn) is linked to lower monthly charges.
- **Tenure vs Total Charges:** The diagonal pattern on the left low scatter plot chart indicates a positive correlation. As tenure increases, total charges generally increase as well. In other words, longer-term customers tend to accumulate higher total charges.

```
PROBLEMS 18   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   QUERY RESULTS        powershell

=====CHURN PERCENTAGE=====
WILL NOT CHURN: 0.00%
WILL CHURN: 0.00%

=====REVENUE SUMMARY=====
Actual Total Revenue: $58,456,608.66
Predicted Total Revenue: $58,456,545.27

Actual Monthly Revenue: $1,604,626.93
Predicted Monthly Revenue: $1,604,624.93
PS C:\Users\User\Desktop\FUTURE_DS_01>
Ln 127, Col 1   Spaces: 4   UTF-8   CRLF   Python   Python 3.14.3
```

- After doing the model and predictions with python, my model predicted that the monthly charges will be at $1,604,624.93 which is 20,058% of the current monthly charges.
- After doing the model and predictions with python, my model predicted that the total revenue of $58,456,608.66 will be made.
- As for the predicted churn and non-churn rates, the model gave me 0% values which is not correct/valid, due to this error, I have to go back and rectify the mistakes I did on the model.